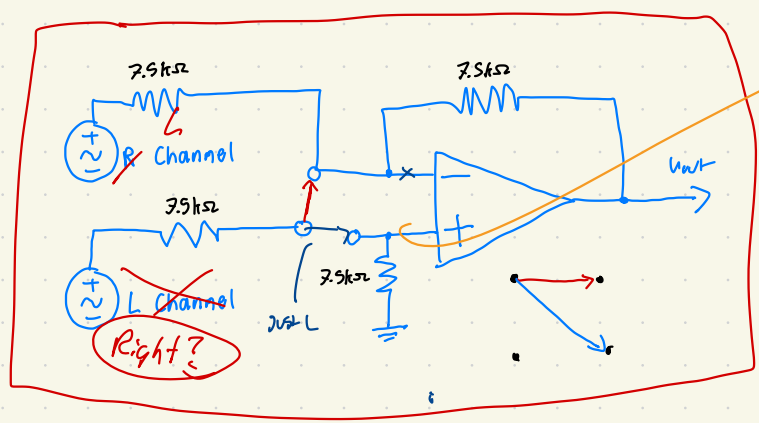


Gain for inverting summing: $-\frac{R_f}{R_i}$ • 7.5kΩ has 1% tolerance

Block 1 Mixer/Karaoke

Gain for subtracting amp: $\frac{R_2}{R_1}(v_2 - v_1)$, with $\frac{R_2}{R_1} = \frac{R_4}{R_3} = \frac{R_f}{R_i}$



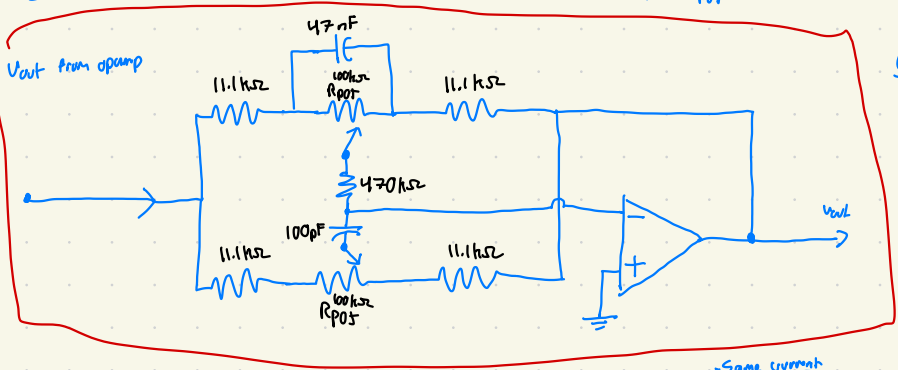
$v_{out} = -v_{in}$

$\frac{7.5}{7.5} \approx 1$

Gain is $1 - (-1) = 2!$

Block 2 Tone Control

Min Gain: $-\frac{R}{R + R_{pot}}$ Max Gain: $-\left(\frac{R + R_{pot}}{R}\right)$



Calculation

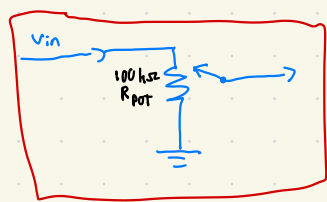
$1 + \frac{100k\Omega}{R} = 10$

Gain is $1 \rightarrow 10$

$R = \frac{100k\Omega}{9} = 11.1k\Omega$

Block 3 Volume Control

Block 4 Volume Display



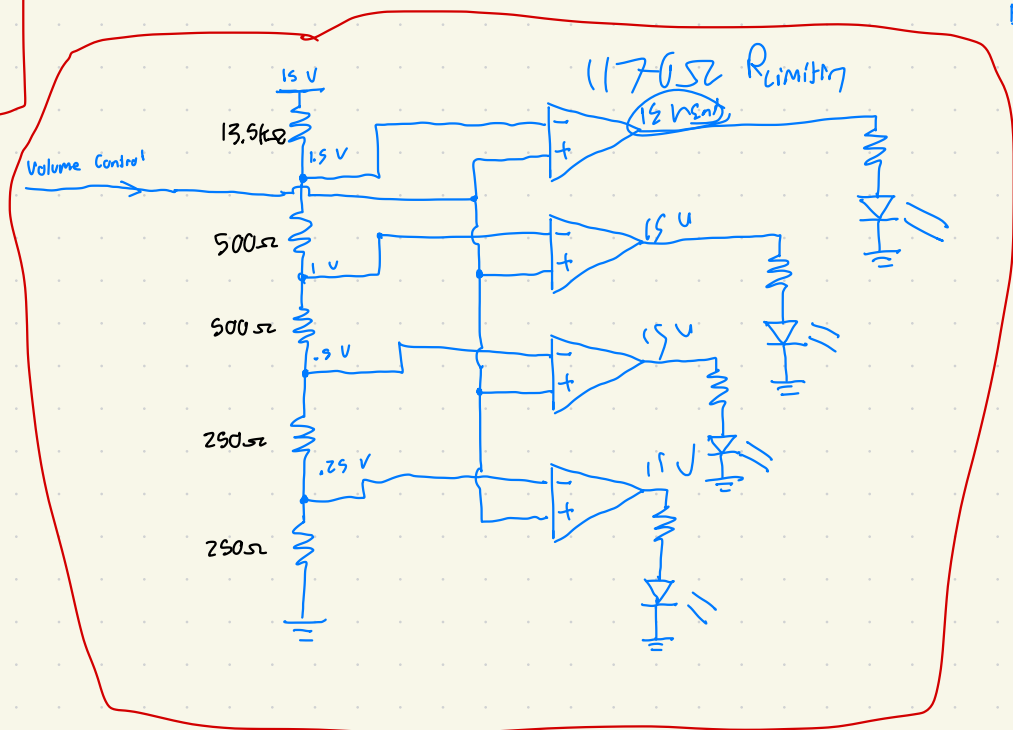
- Same current
for 1.5 V:
V drop: $1.5 - 1.3 = 13.5$
 $\frac{13.5V}{1\Omega} = 13.5k\Omega$
using pot

For 1V:
 $v_{drop}: 1.5 - 1.3 = 13.5$
 $\frac{13.5}{1} = 13.5k\Omega$

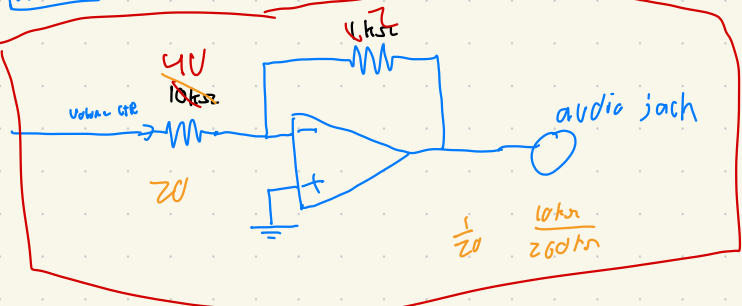
For 0.5V:
 $1.5 - 1.3 = 13.5$
 $\frac{13.5}{1} = 13.5k\Omega$

For 0.25V:
 $1.5 - 1.3 = 13.5$
 $\frac{13.5}{1} = 13.5k\Omega$

For GND:
 $\frac{13.5}{1} = 13.5k\Omega$



Block 5 Attenuator and Output Buffer



Gain: $\frac{R_f}{R_i} = 1$ (bc gain can be 10)

$\frac{1k}{10k} = 0.1$

needs to be 1/20, not 1/10
cs more gain is 20

$\frac{2}{40} = \frac{1}{20}$

Block 4 Volume Display

- Same current
for 1.5 V:
 $1 - x = 1.5$
 $x = 3.5$
 $3.5 = R$

For 1V:
 $1 - 1 = .5$
 $.5 = R$
 $500\Omega = R$

For 0.5 V:
 $1 - .5 = .5$
 $.5 = R$
 500Ω

For 0.25 V:
 $.5 - .25 = .25$
 $.25 = R$
 250Ω

$.25$
 250Ω

$1 - 3.3$
 $.01$
 $530\Omega R_L$

